Stock forecasts based on the XGBoost model

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Abstract

In this study, we utilize the XGBoost model to predict stock prices, addressing the urgent need for reliable investment strategies in the volatile stock market. We review traditional and modern prediction methods, emphasizing machine learning advancements. Data preprocessing and feature engineering techniques are detailed, followed by the implementation of the XGBoost model, enhanced with Support Vector Machine (SVM) features to reduce overfitting. Experimental results indicate significant improvements in prediction accuracy and speed. The study highlights the model's strengths and limitations, suggesting future enhancements for better handling of sudden market fluctuations.

CCS Concepts

• **General and reference** → Document types; Computing standards, RFCs and guidelines.

Keywords

Prediction of Stock Price, XGBoost Model, Machine Learning, Feature Engineering, Time Series Data, Support Vector Machine (SVM)

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1 INTRODUCTION

Society's perpetual growth and the enhancement of living standards have now endowed people with a surplus of funds to invest in, having addressed fundamental requirements.As a result, more and more people are turning their focus to investing in stocks, supplying monetary backing for the long-term growth of the stock market. However, in the complex and ever-changing stock market, identifying the optimal stocks and ensuring positive growth of funds has become an urgent issue. This challenge is faced not only by many shareholders but also by scholars focused on stock price prediction.

Many factors have an effect on stock prices, which are highly unpredictable over time.Economic, political and the company's own operational conditions are the major external influences. Since the establishment of the stock market, researchers have employed a range of methodologies to study stock price fluctuations. From an economic perspective, investors generally use traditional fundamental analysis, technical analysis and evolutionary analysis to predict stock prices. Though theoretical, these customary examination techniques cannot completely demonstrate the relationship between the facts.As mathematical statistics and machine learning progress, more and more people are utilizing modern forecasting techniques for stock forecasting, such as neural network prediction, decision tree prediction, and support vector machine prediction.Prediction of logistic regression and deep learning.

Quantitative analysis of stocks, utilizing the expertise of mathematics, economics, and computer science, is a method of research that can forecast stock trends based on past data. The application of machine learning algorithms to stock market analysis is on the rise due to the continual advancement of machine learning technology.Algorithms such as support vector machines (SVM), decision trees, random forests, AdaBoost, and gradient boosting decision trees (GBDT) are all included. By employing these algorithms, researchers can extract valuable insights from vast amounts of stock trading data to more accurately predict future stock prices and movements.

Stock data is a type of time-series data characterized by large volumes, numerous indicators, and nonlinearity. Consequently, this paper utilizes the XGBoost model, known for its robust multivariable processing capabilities and rapid operation speed. The XGBoost model, an open-source machine learning project developed by Tianqi Chen[1] in 2016, has demonstrated its advantages in computational complexity, running speed, and accuracy, as evidenced in the literature. XGBoost effectively implements the GBDT algorithm. While GBDT can significantly enhance stock prediction in time-series data analysis, its detection rate is relatively slow. Employing the XGBoost model, stock prediction is achieved with both swiftness and precision.Results from the experiment demonstrate that XGBoost not only boosts prediction accuracy but also significantly boosts prediction velocity. In recent years, XGBoost has performed exceptionally well in various machine learning competitions. The XGBoost model's prediction accuracy is further augmented by the use of the Support Vector Machine (SVM) model. Experimental results indicate that the index expansion provided by SVM effectively reduces the overfitting of the XGBoost model, significantly improving its predictive performance.

Forecasting stock prices has been a highly sought-after subject in the financial sector, and its potential for high economic returns has drawn the interest of numerous researchers both domestically and internationally. Box[2] and Jenkins' ARIMA (autoregressive integral sliding average) model, a pioneering use of computers in stock forecasting, was first proposed in the 1960s; and it was

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further developed in the 1990s. Saad[3]et al. used neural networks for stock trend prediction, 2010 J.G. Refenes[4]et al. put artificial neural networks in financial markets for volatility prediction, 2017 L. Bao[5]et al. explored the application of convolutional neural network (CNN) in multivariable time series stock market prediction, 2018 S. Fischer[6]et al. Using the long and short-term memory network (LSTM) for financial time series prediction, 2020 S. Makridakis[7]Putting learning models such as LSTM, GRU and CNN into financial time series prediction, Makridakis et al. while in 2021, D. Zhang[8] et al. In 2022, Y. Li, Y. [9] conducted a study to explore the use of neural networks and the BP algorithm in the categorization and forecasting of stock price trends.A deep learning integration model, based on stock prices and news for forecasting stocks, has been proposed by Pan. Experiments demonstrate that the proposed model outperforms conventional models in prediction accuracy. In 2023, Ramos-Perez[10] et al. discussed the application of ARIMA model, wavelet transform and GARCH model in stock market volatility prediction, showing the advantages of this hybrid model in improving prediction accuracy. In 2024, Siyuan Zhu[11] predicted the US stock market index combined with dynamic ARIMA model and long-term and short-term memory (LSTM) model. In 2017, Zhang Li[12] employed the Deep Neural Network (DNN) model to forecast the RMB exchange rate in China. Then, in 2018, Liu Tao[13] proposed a multivariate time series prediction technique based on Convolutional Neural Network (CNN).For 2020, Y's[14] stock market forecasts shall be applied.Wang et al.Huang investigated the efficacy of LSTM and GRU networks in the financial markets' volatility in 2021, Chen, M., & Huang, W.[15]utilizing a hybrid model of ARIMA and XGBoost to forecast stock market volatility over time series data. In 2023, Zou Jie[16] et al. based on the SA-BiGRU model. In 2024, Sun Chenyu[17] et al. based on the stock market prediction model of CEEMDAN and LSTM-Attention, and this model is better than the existing model.

At the same time, although the XGBoost model has only been proposed for eight years, it has been widely used because of its advantages of strong and stable processing ability and fast operation speed. In 2018, Yuan Kun[18] used the XGBoost model to predict the use of mobile coupons, In 2019, Li Huixuan[19]et al. combined the XGBoost model and SVR, In 2021, Wang Haoyu[20]et al. used the spatial interpolation model of precipitation in China developed based on Xgboost + GWR algorithm; These studies have all deeply inspired me deeply, So I chose the XGBoost model.In 2022, Xu Li[21] et al. evaluated the stock price prediction method based on LSTM and news emotion. In 2024, Zhang Xuntao[22] et al evaluated the stock trend analysis based on the LSTM model selected based on XGBOOST characteristics

This article is structured as follows: the first section introduces the XGBoost model, its application background, significance in stock prediction, and reviews the relevant literature."In the second paragraph, namely,the data section, the processes of data acquisition, preprocessing,and feature engineering are described. Then, in the third paragraph, the experimental design and result analysis are elaborated, showing the performance of XGBoost model in stock prediction. We then debate the benefits and drawbacks of the model, its restrictions, and suggest the path to betterment. In conclusion, we present the research findings and suggest potential avenues for further exploration in the fourth paragraph.The text also contains all the referenced sources.

2 BACKGROUNDS

XGBoost As a Boosting model, proposed by Dr. Chen Tianqi at the University of Washington, he has attracted wide attention for his outstanding performance in the machine learning challenge. Compared to the increasingly popular deep neural networks, XGBoost demonstrates greater capabilities and higher interpretability in processing tabular data. XGBoost It has the significant advantages of easy adjustment of parameters and invariance of input data, which is identical to other Boosting models and is an additive model. XGBoost A series of binary tree classifiers are trained, and each classifier has at least two leaf nodes. The regression tree's prediction results are the weights of the leaf nodes, while the model's ultimate prediction score is the weight of the leaf nodes of multiple classifiers.

XGBoost like the other Boosting models, they are all additive models. Popular understanding: assuming that the base model is a CART regression tree model ('booster' = 'gbtree'), XGBoost trained a series of binary tree classifiers, and each classifier has at least two leaf nodes. The regression tree's prediction result is the leaf node weight, and the model's ultimate prediction result score is the combination of the leaf node weights from multiple classifiers. The subsequent processing is different according to the scores of the task. For example, the final score of the task will be mapping to the value of [0,1] using Sigmoid function to complete the prediction task (the specific process and the conversion mode of multi-classification and regression will be explained in detail in the subsequent sections).

The mathematical definition of the regression tree is given and the parameters of XGBoost are explained. This section, without not stated, are quoted from reference [1].

For n rows, data sets with m indicators and labels y D= {(xi, yi)}, where | D | =n, xiRm, yiR. Mapping the data to the leaf node, F= {f (x) = ω q (x)}, yields q: RmT, with T representing the label set of the leaf child nodes and ω i the score on the leaf node i, forming the set of the regression tree. The predictive value of the additive regression tree model ϕ for a K tree is defined as

$$\hat{y}_i = \varphi\left(x_i\right) = \sum_{k=1}^{K} f_k\left(x_i\right), f_k \in F \tag{1}$$

To describe the fitting effect of the model, the loss function L is defined as follows

$$L\left(\varphi\right) = \sum_i l\left(\hat{y}_i, y_i\right) + \sum_k \Omega\left(f_k\right) \tag{2}$$

Among

$$\Omega\left(f\right) = \gamma\left|T\right| = \frac{1}{2}\lambda\omega_2^2 \tag{3}$$

Here is to represent a function of the residual size between the predicted value and the actual value, and Ω is to represent the model complexity

A function of degree.

XGBoost's concept is to continually augment each iteration with a tree, thus keeping the anticipated value near the real one; thus,

the aim of the tth iteration is to reduce the following functions.

$$L^{(t)} = \sum_{i}^{n} l\left(y_i, \hat{y}_i^{(t-1)} + f_t\left(x_i\right)\right) + \Omega\left(f_t\right) \tag{4}$$

After a series of derivations, IL is the set of samples divided into the left child, similarly I, R is the set of the right child, I = ILI

R is the label set of the paternal node. Then the loss function caused by a single split changes as follow

$$L_{\text{split}} = \frac{1}{2}\left(\frac{\left(\sum i \in I_L g_i\right)^2}{\sum i \in I_L h_i + \lambda} + \frac{\left(\sum i \in I_L g_i\right)^2}{\sum i \in I_R h_i + \lambda} - \frac{\left(\sum i \in I_R g_i\right)^2}{\sum i \in I h_i + \lambda}\right) - \gamma \tag{5}$$

So far, we have completed the basic introduction of XGBoost

Performance in heterogeneous data processing, and it is validated experimentally.

3 RONSTRUCTION

XGBoost The construction of the model is no different from that of other data models. It still includes the five steps of data acquisition, data preprocessing, data division, model construction, evaluation and evaluation.

Acquiring stock data sets is a pivotal step in the modeling process, for the quality of the data will determine the model's result; this is the initial step. If the data quality is poor, then using a good model will not get very good results. In general, good data should first have no or few missing values and outliers, and the indicators of the data should have a high correlation with the predictive dependent variables. Using such data will greatly improve the model and the effect of the model.

Second, the data preprocessing was performed. In actual modeling, the data obtained belongs to dirty data, which needs to be processed before modeling. A plethora of data preprocessing techniques exist, such as data cleaning, integration, transformation, and reduction, to name a few. What type of data preprocessing should be employed, based on the characteristics and research issues? This paper mainly utilized data cleaning to fill in missing value.By detecting the missing value of the original data, outlier detection and data transformation of data standardization can be employed to ascertain if the original data is complete.By utilizing the sklearn StandardScaler module in the experiment, outliers can be identified to ascertain if the original data is accurate.

After data preprocessing, it is necessary to build more indicators affecting the predictive dependent variables based on the original indicators, which is the feature engineering in the modeling process mentioned above. This paper constructs the stock price change and rise and fall based on the opening, closing, highest, lowest, and trading volume. Some index data are shown in (Figure 1).

Finally, in constructing stock price and trend prediction model, the XGBoost model proposed by Dr and D, Due to its excellent performance on large-scale, multi-indicator nonlinear datasets. We augmented the model's learning capacity by augmenting its feature set with the support vector machine (SVM) during the pre-planning stage. Subsequently, an important process —— parameter optimization is used to improve our prediction performance and make a finishing point. During modeling, the model can be directly imported through Python's xgboost library.

	Date	Open	Close	High	Low	Volume	Amount	\
0	2017-03-01	9.49	9.49	9.55	9.47	346994	3.301580e+08	
1	2017-03-02	9.51	9.43	9.54	9.42	403629	3.823959e+08	
2	2017-03-03	9.41	9.40	9.43	9.36	342655	3.219525e+08	
3	2017-03-06	9.40	9.45	9.46	9.39	404511	3.812123e+08	
4	2017-03-07	9.44	9.45	9.46	9.40	294673	2.777474e+08	
...	...	...	...	...	...	...	...	
1755	2024-05-22	11.56	11.56	11.74	11.46	2115531	2.458449e+09	
1756	2024-05-23	11.53	11.40	11.59	11.37	1841623	2.110799e+09	
1757	2024-05-24	11.37	11.31	11.49	11.30	1398276	1.593330e+09	
1758	2024-05-27	11.31	11.51	11.53	11.31	1454361	1.663272e+09	
1759	2024-05-28	11.50	11.40	11.58	11.36	1204323	1.377107e+09	

	Amplitude	Change Percentage	Change Amount	Turnover Rate
0	0.84	0.11	0.01	0.21
1	1.26	-0.63	-0.06	0.24
2	0.74	-0.32	-0.03	0.20
3	0.74	0.53	0.05	0.24
4	0.63	0.00	0.00	0.17
...	...	...	...	...
1755	2.42	0.09	0.01	1.09
1756	1.90	-1.38	-0.16	0.95
1757	1.67	-0.79	-0.09	0.72
1758	1.95	1.77	0.20	0.75
1759	1.91	-0.96	-0.11	0.62

[1760 rows x 11 columns]

Figure 1: The resource of stock.

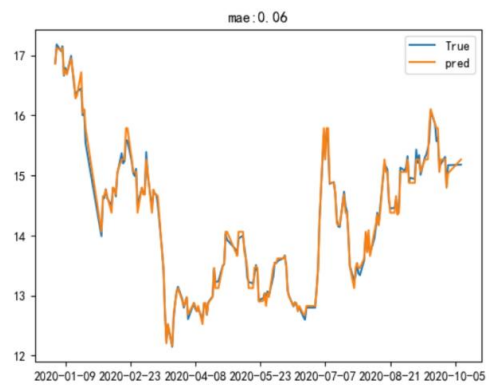

Figure 2: Result of prediction.

4 RESULTS

We chose stocks with the ticker code 000001.SZ for daily trading data from January 1,2018 to December 31,2023 (Figure 2). Data were acquired through the Tushare library and saved as a CSV file. Indexing the date, we then converted it to a time series format for further analysis and modeling.The following fields are included in the data obtained: opening price, closing price, highest price, lowest price, and trading volume, among others.

To capture the dependencies in the time series data, we constructed lag features with the transaction data from the first 1 to 6 days as input features for the model. The closing price, opening price, high price, low price, and trading volume are all characteristics of this. By constructing these lag features, we are able to better use the historical data and improve the predictive power of the model.

On the basis of the existing model, we can further reduce the mean absolute error (MAE) by using the Stacking method (stacking model). Stacking Is an integrated learning method that improves the overall performance of the model by combining the prediction results of multiple base learners (Figure 3.)

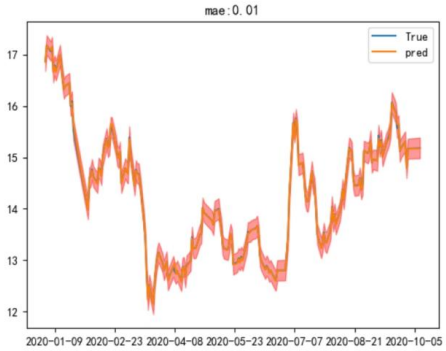

Figure 3: Reduce the mae.

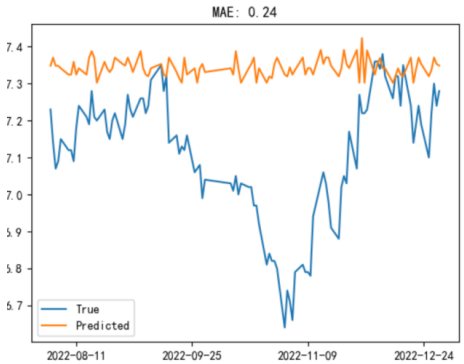

Figure 4: The accident result.

When we found that mae rose after stacking, it can be seen that the predictive ability of this model still needs to be improved. Therefore, we can further adjust the parameters of each model with Stacking aggregation through GridSearchCV to reduce the mae level in

However, when applying our machine learning model to the stock forecast of SPD Bank, we found that the prediction results of the model were significantly different from the actual data. (Figure 4). After analysis, this anomaly was due to the annual report issued by SPD Bank during this period, which resulted in the stock price fluctuations completely different from the forecast results. This finding indicates that when using the XGBoost model and the RFECV (recursive feature elimination and cross-validation) feature selection method for stock prediction, the drastic fluctuations caused by external factors can significantly affect the prediction accuracy of the model.

This suggests that while the combination of XGBoost and RFECV provides more accurate forecasts in most cases, the forecast performance of the model may be severely affected in the face of unexpected external events (such as earnings release, policy changes, etc.)

5 SUMMMARY

We constructed a model for predicting stock closing prices by employing the XGBoost model and the RFECV (Recursive Feature Elimination with Cross-Validation) feature selection technique in this experiment. Experimental findings demonstrate that the model is capable of accurately forecasting the stock closing price in the majority of cases, with a satisfactory overall performance. However, the model still has some errors. Specifically, the RFECV feature selection method helps us to screen the most influential features on the stock closing price, thus reducing the redundant features and improving the training efficiency and prediction accuracy of the model. The XGBoost model further enhances its performance due to its advantages in handling nonlinear relationships and preventing overfitting. The model's inability to anticipate and capture the machine is exacerbated by the high volatility of the stock market, resulting in unexpected external factors such as financial and policy announcements, as well as the market's excessive volatility.The high volatility area necessitates an enhancement of its performance. In this regard, more technical indicators, macroeconomic data and news sentiment analysis are introduced to enrich the input information of the model, and more data can also be introduced to strengthen the stability and prediction ability of the model, so as to make it better generalized

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